

The Sum and Difference Identities

The Sine of the Sum or Difference of Two Angle Measures

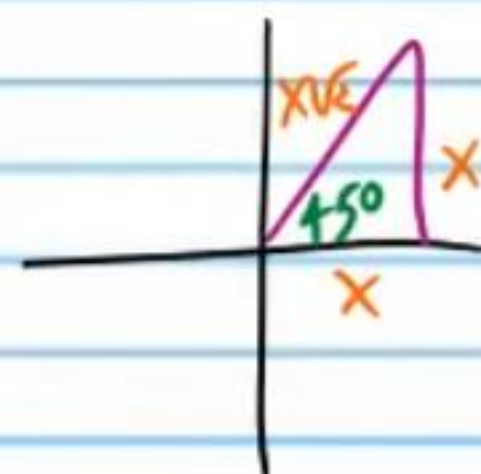
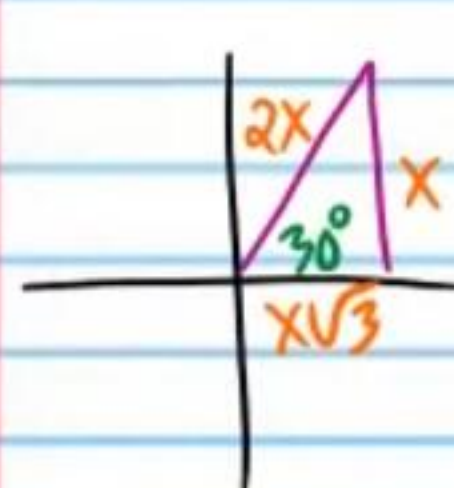
$$\sin(A \pm B) = \sin A \cos B \pm \sin B \cos A$$

The Cosine of the Sum or Difference of Two Angle Measures

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

Find the values of the following expressions *without a calculator*.

• ① $\sin 75^\circ = \sin(30^\circ + 45^\circ) = \sin 30^\circ \cos 45^\circ + \sin 45^\circ \cos 30^\circ$
 $= \frac{1}{2} \left(\frac{1}{\sqrt{2}} \right) + \frac{1}{\sqrt{2}} \left(\frac{\sqrt{3}}{2} \right)$



$$= \frac{1}{2\sqrt{2}} + \frac{\sqrt{3}}{2\sqrt{2}}$$

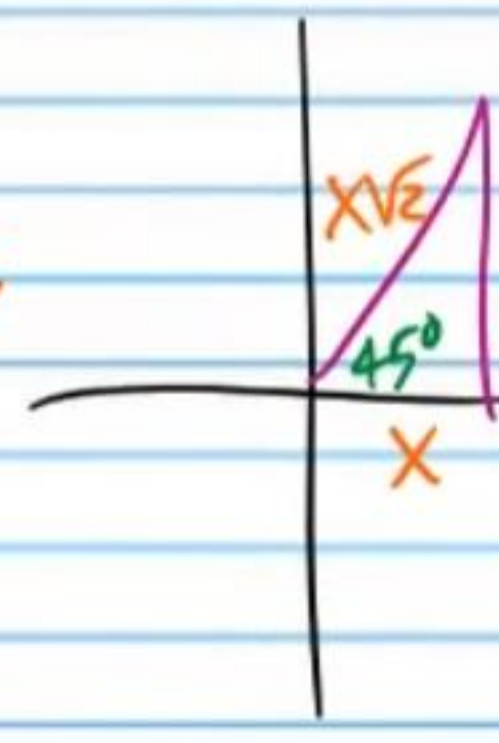
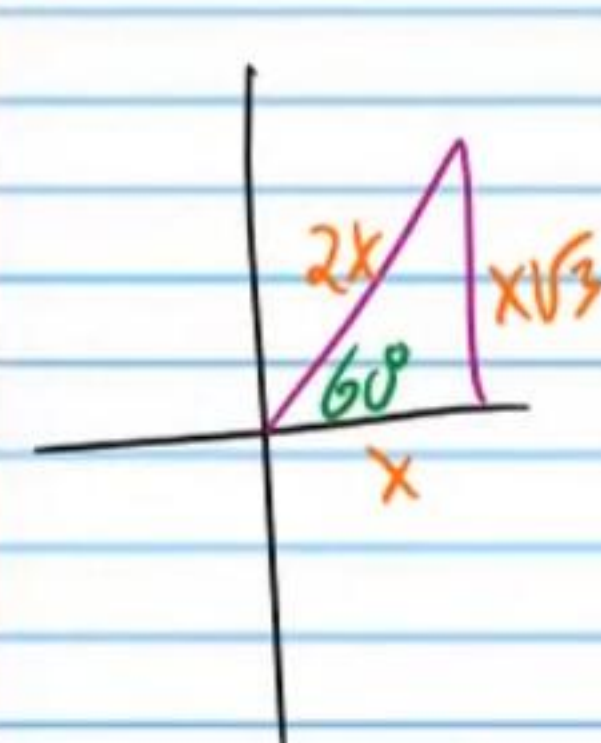
$$= \frac{1 + \sqrt{3}}{2\sqrt{2}}$$

$\sqrt{a} \cdot \sqrt{b} = \sqrt{ab}$

$$= \frac{(1 + \sqrt{3})\sqrt{2}}{(2\sqrt{2})\sqrt{2}}$$

$$= \boxed{\frac{\sqrt{2} + \sqrt{6}}{4}}$$

② $\cos 105^\circ = \cos(60^\circ + 45^\circ) = \cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ$
 $= \frac{1}{2} \left(\frac{1}{\sqrt{2}} \right) - \frac{\sqrt{3}}{2} \left(\frac{1}{\sqrt{2}} \right)$



$$= \frac{1}{2\sqrt{2}} - \frac{\sqrt{3}}{2\sqrt{2}}$$

$$= \frac{1 - \sqrt{3}}{2\sqrt{2}}$$

$$= \boxed{\frac{\sqrt{2} - \sqrt{6}}{4}}$$

• ③ $\cos 75^\circ = \cos(30^\circ + 45^\circ) = \cos 30^\circ \cos 45^\circ - \sin 30^\circ \sin 45^\circ$
 $= \frac{\sqrt{3}}{2} \left(\frac{1}{\sqrt{2}} \right) - \frac{1}{2} \left(\frac{1}{\sqrt{2}} \right)$

$$= \frac{\sqrt{3}}{2\sqrt{2}} - \frac{1}{2\sqrt{2}}$$

$$= \frac{\sqrt{3} - 1}{2\sqrt{2}}$$

$$= \boxed{\frac{\sqrt{6} - \sqrt{2}}{4}}$$

$$\textcircled{4} \sin 105^\circ = \sin(60^\circ + 45^\circ) = \sin 60^\circ \cos 45^\circ + \sin 45^\circ \cos 60^\circ$$

$$= \frac{\sqrt{3}}{2} \left(\frac{1}{\sqrt{2}} \right) + \frac{1}{\sqrt{2}} \left(\frac{1}{2} \right)$$

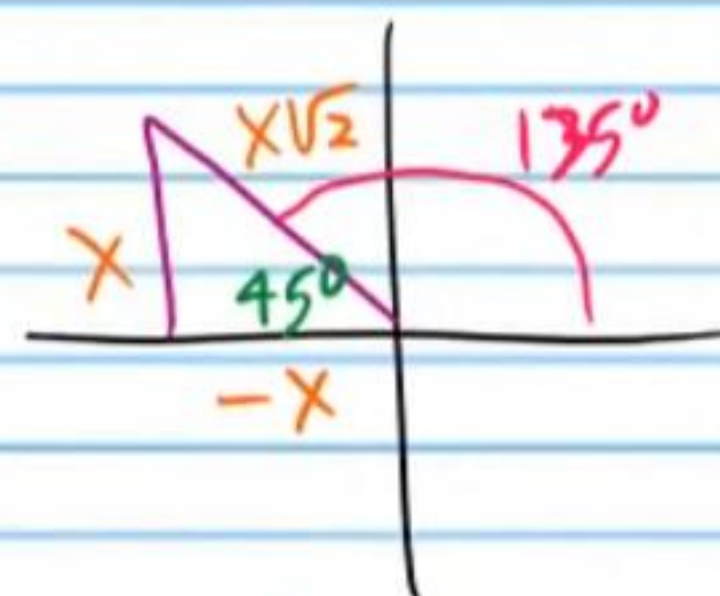
$$= \frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}}$$

$$= \frac{\sqrt{3}+1}{2\sqrt{2}}$$

$$= \boxed{\frac{\sqrt{6}+\sqrt{2}}{4}}$$

$$\bullet \textcircled{5} \sin 195^\circ = \sin(60^\circ + 135^\circ) = \sin 60^\circ \cos 135^\circ + \sin 135^\circ \cos 60^\circ$$

$$= \frac{\sqrt{3}}{2} \left(-\frac{1}{\sqrt{2}} \right) + \frac{1}{\sqrt{2}} \left(\frac{1}{2} \right)$$



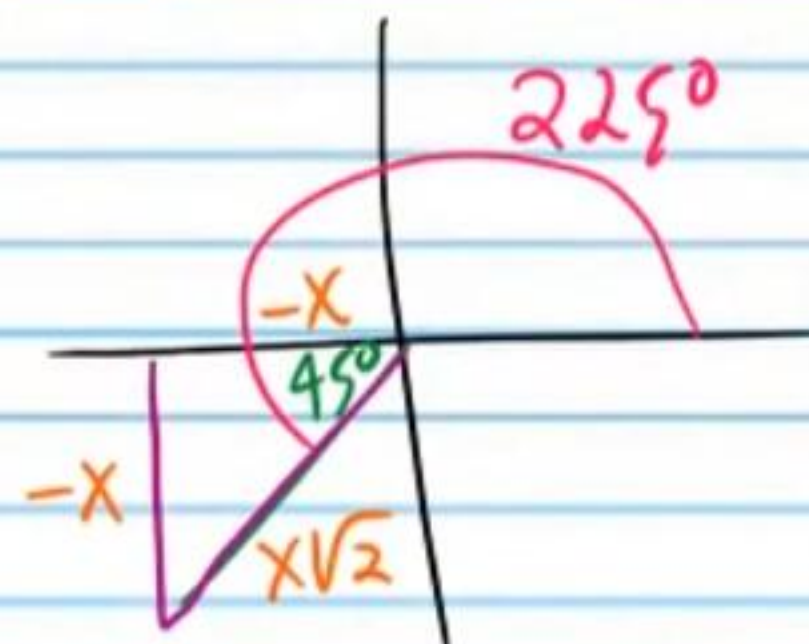
$$= -\frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}}$$

$$= \frac{-\sqrt{3}+1}{2\sqrt{2}}$$

$$= \boxed{\frac{-\sqrt{6}+\sqrt{2}}{4}}$$

$$\textcircled{6} \cos 285^\circ = \cos(60^\circ + 225^\circ) = \cos 60^\circ \cos 225^\circ - \sin 60^\circ \sin 225^\circ$$

$$= \frac{1}{2} \left(-\frac{1}{\sqrt{2}} \right) - \frac{\sqrt{3}}{2} \left(-\frac{1}{\sqrt{2}} \right)$$



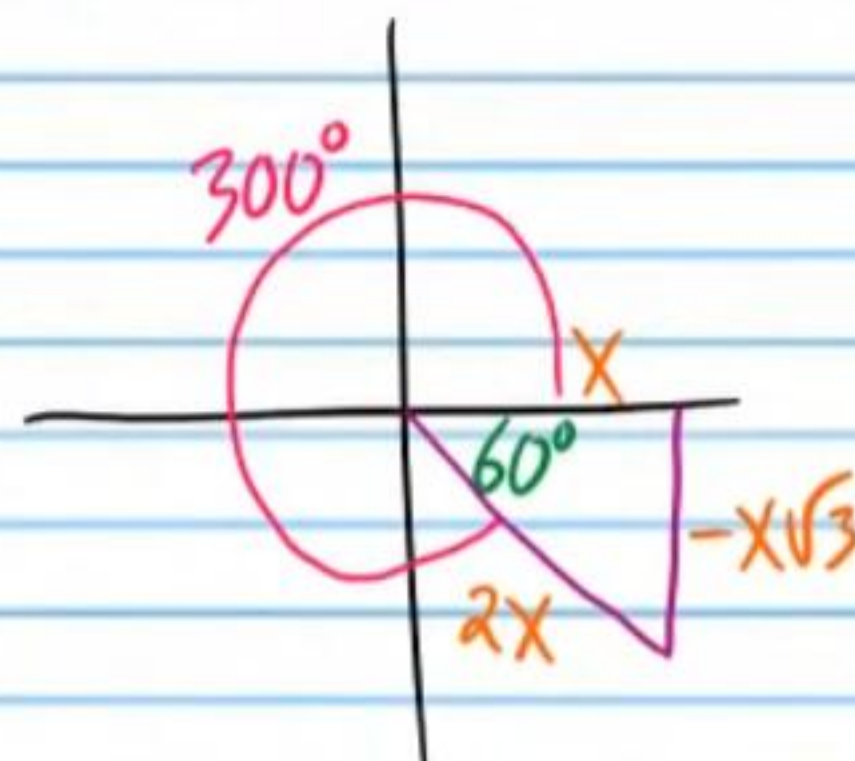
$$= -\frac{1}{2\sqrt{2}} + \frac{\sqrt{3}}{2\sqrt{2}}$$

$$= \frac{-1+\sqrt{3}}{2\sqrt{2}}$$

$$= \boxed{\frac{-\sqrt{2}+\sqrt{6}}{4}}$$

$$\bullet \textcircled{7} \sin 345^\circ = \sin(300^\circ + 45^\circ) = \sin 300^\circ \cos 45^\circ + \sin 45^\circ \cos 300^\circ$$

$$= -\frac{\sqrt{3}}{2} \left(\frac{1}{\sqrt{2}} \right) + \frac{1}{\sqrt{2}} \left(\frac{1}{2} \right)$$



$$= -\frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}}$$

$$= \frac{-\sqrt{3}+1}{2\sqrt{2}}$$

$$= \boxed{\frac{-\sqrt{6}+\sqrt{2}}{4}}$$

$$\begin{aligned}
 \textcircled{8} \cos 255^\circ &= \cos(30^\circ + 225^\circ) = \cos 30^\circ \cos 225^\circ - \sin 30^\circ \sin 225^\circ \\
 &= \frac{\sqrt{3}}{2} \left(-\frac{1}{\sqrt{2}}\right) - \frac{1}{2} \left(-\frac{1}{\sqrt{2}}\right) \\
 &= -\frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} \\
 &= \frac{-\sqrt{3} + 1}{2\sqrt{2}} \\
 &= \boxed{\frac{-\sqrt{6} + \sqrt{2}}{4}}
 \end{aligned}$$

$$\begin{aligned}
 \bullet \textcircled{9} \sin 15^\circ &= \sin(45^\circ - 30^\circ) = \sin 45^\circ \cos 30^\circ - \sin 30^\circ \cos 45^\circ \\
 &= \frac{1}{\sqrt{2}} \left(\frac{\sqrt{3}}{2}\right) - \frac{1}{2} \left(\frac{1}{\sqrt{2}}\right) \\
 &= \frac{\sqrt{3}}{2\sqrt{2}} - \frac{1}{2\sqrt{2}} \\
 &= \frac{\sqrt{3} - 1}{2\sqrt{2}} \\
 &= \boxed{\frac{\sqrt{6} - \sqrt{2}}{4}}
 \end{aligned}$$

$$\begin{aligned}
 \bullet \textcircled{10} \cos 15^\circ &= \cos(60^\circ - 45^\circ) = \cos 60^\circ \cos 45^\circ + \sin 60^\circ \sin 45^\circ \\
 &= \frac{1}{2} \left(\frac{1}{\sqrt{2}}\right) + \frac{\sqrt{3}}{2} \left(\frac{1}{\sqrt{2}}\right) \\
 &= \frac{1}{2\sqrt{2}} + \frac{\sqrt{3}}{2\sqrt{2}} \\
 &= \frac{1 + \sqrt{3}}{2\sqrt{2}} \\
 &= \boxed{\frac{\sqrt{2} + \sqrt{6}}{4}}
 \end{aligned}$$

Simplify the following expressions. Write exact expressions.
In other words, you do not have to use a calculator.

- $\textcircled{11} \sin x \cos 2x + \sin 2x \cos x = \sin(x + 2x) = \boxed{\sin(3x)}$
- $\textcircled{12} \cos 7\theta \cos 2\theta + \sin 7\theta \sin 2\theta = \cos(7\theta - 2\theta) = \boxed{\cos(5\theta)}$
- $\textcircled{13} \sin 18^\circ \cos 24^\circ - \sin 24^\circ \cos 18^\circ = \sin(18^\circ - 24^\circ) = \boxed{\sin(-6^\circ)}$
- $\textcircled{14} \cos 37^\circ \cos 22^\circ - \sin 37^\circ \sin 22^\circ = \cos(37^\circ + 22^\circ) = \boxed{\cos 59^\circ}$
- $\textcircled{15} \sin 3B \cos 8B - \sin 8B \cos 3B = \sin(3B - 8B) = \boxed{\sin(-5B)}$

$$(16) \cos 6W \cos W - \sin 6W \sin W = \cos(6W+W) = \boxed{\cos(7W)}$$

$$(17) \sin 100^\circ \cos 10^\circ + \sin 10^\circ \cos 100^\circ = \sin(100^\circ+10^\circ) = \boxed{\sin 110^\circ}$$

$$(18) \cos 75^\circ \cos 25^\circ + \sin 75^\circ \sin 25^\circ = \cos(75^\circ-25^\circ) = \boxed{\cos 50^\circ}$$

Prove that the following equations are true by converting the left side to the right.

$$\bullet (19) \sin(x+y) - \sin(x-y) = 2 \sin y \cos x$$

$$\sin x \cos y + \sin y \cos x - (\sin x \cos y - \sin y \cos x) =$$

$$\sin x \cos y + \sin y \cos x - \sin x \cos y + \sin y \cos x =$$

$$2 \sin y \cos x =$$

$$(20) \cos 3x + \sin x \sin 2x = \cos x \cos 2x$$

$$\cos(2x+x) + \sin x \sin 2x =$$

$$\cos 2x \cos x - \sin 2x \sin x + \sin x \sin 2x =$$

$$\cos 2x \cos x =$$

$$(21) \cos 50^\circ = \cos 130^\circ \cos 80^\circ + \sin 130^\circ \sin 80^\circ$$

$$\cos(130^\circ-80^\circ) =$$

$$\cos 130^\circ \cos 80^\circ + \sin 130^\circ \sin 80^\circ =$$

$$\bullet (22) \cos(\alpha-\beta) + \cos(\alpha+\beta) = 2 \cos \alpha \cos \beta$$

$$\cos \alpha \cos \beta + \sin \alpha \sin \beta + \cos \alpha \cos \beta - \sin \alpha \sin \beta =$$

$$2 \cos \alpha \cos \beta =$$

$$(23) \sin 5\theta - \sin 3\theta \cos 2\theta = \sin 2\theta \cos 3\theta$$

$$\sin(3\theta+2\theta) - \sin 3\theta \cos 2\theta =$$

$$\sin 3\theta \cos 2\theta + \sin 2\theta \cos 3\theta - \sin 3\theta \cos 2\theta =$$

$$\sin 2\theta \cos 3\theta =$$

$$(24) \sin 253^\circ = \sin 260^\circ \cos 7^\circ - \sin 7^\circ \cos 260^\circ$$

$$\sin(260^\circ-7^\circ) =$$

$$\sin 260^\circ \cos 7^\circ - \sin 7^\circ \cos 260^\circ =$$

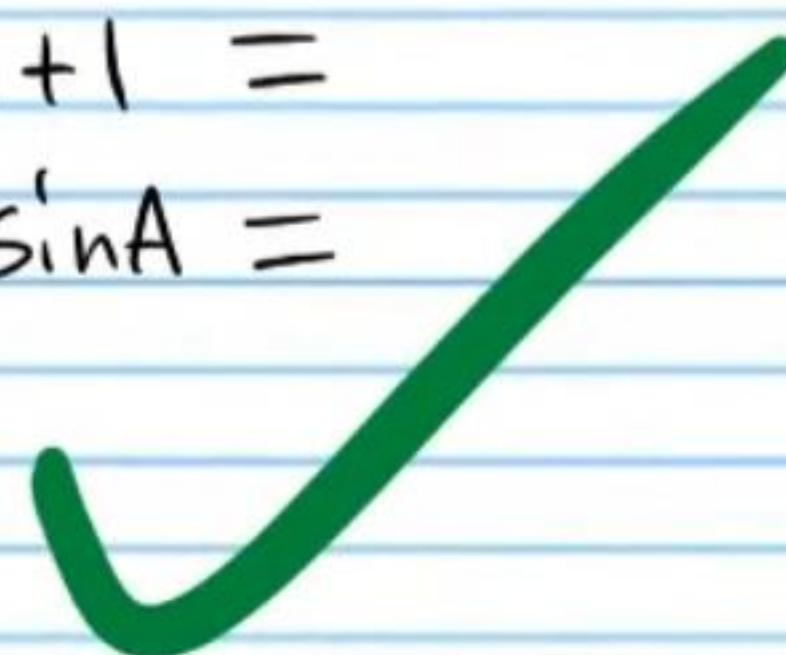
• (25) $\sin(A+180^\circ)+1=1-\sin A$

$$\sin A \cos 180^\circ + \sin 180^\circ \cos A + 1 =$$

$$(\sin A)(-1) + (0)\cos A + 1 =$$

$$-\sin A + 0 + 1 =$$

$$-\sin A + 1 =$$

$$1 - \sin A =$$


(26) $\cos(A-90^\circ) - \sin A = 0$

$$\cos A \cos 90^\circ + \sin A \sin 90^\circ - \sin A =$$

$$(\cos A)(0) + (\sin A)(1) - \sin A =$$

$$0 + \sin A - \sin A =$$

$$0 + 0 =$$

$$0 =$$


• (27) $\cos B + \cos(B+270^\circ) = \sin B + \cos B$

$$\cos B + \cos B \cos 270^\circ - \sin B \sin 270^\circ =$$

$$\cos B + (\cos B)(0) - (\sin B)(-1) =$$

$$\cos B + 0 + \sin B =$$

$$\sin B + \cos B =$$


(28) $\sin(X-360^\circ) = \sin X$

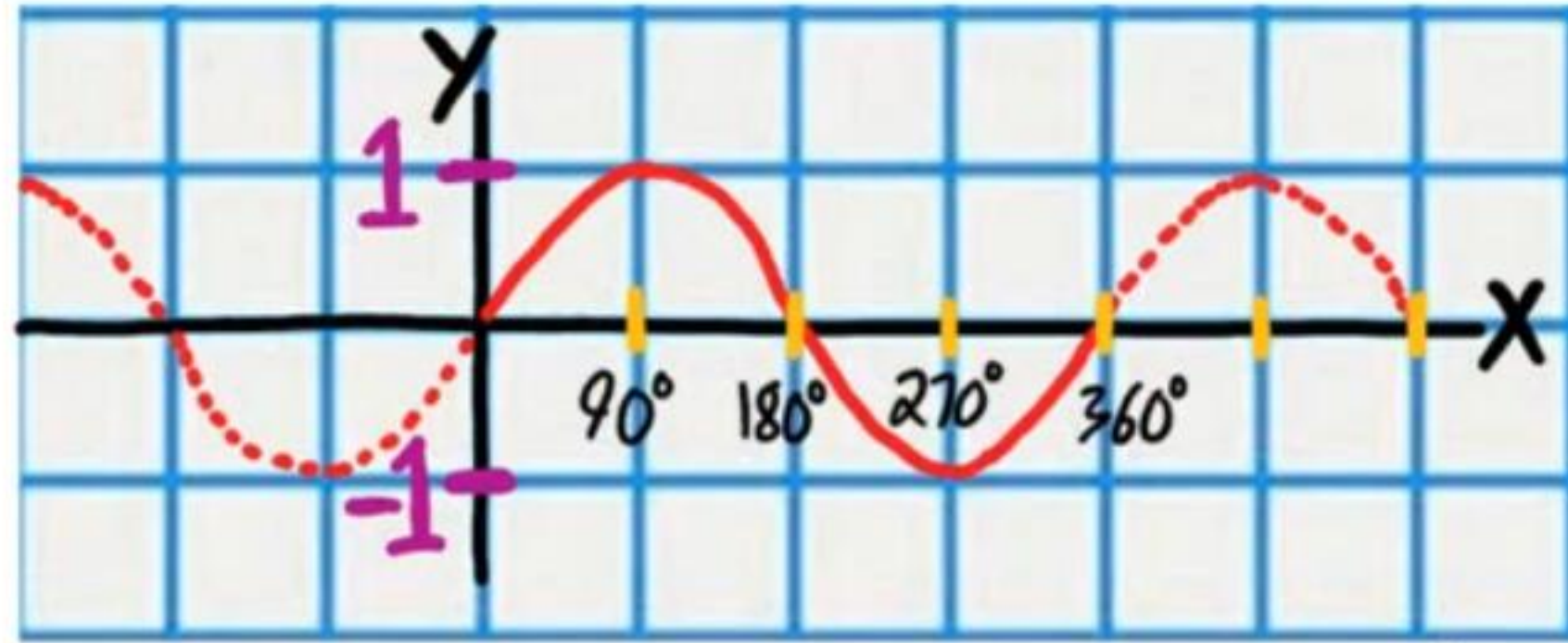
$$\sin X \cos 360^\circ - \sin 360^\circ \cos X =$$

$$(\sin X)(1) - (0)(\cos X) =$$

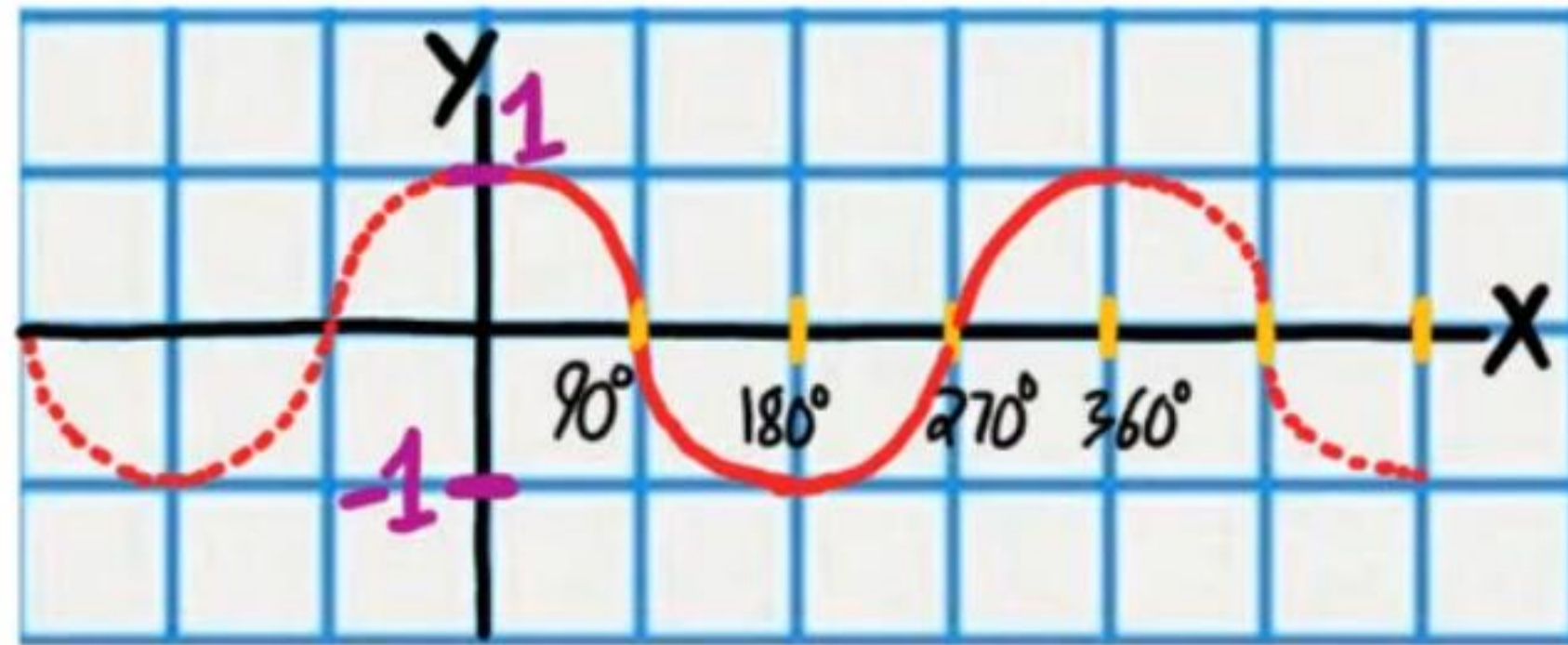
$$\sin X - 0 =$$

$$\sin X =$$

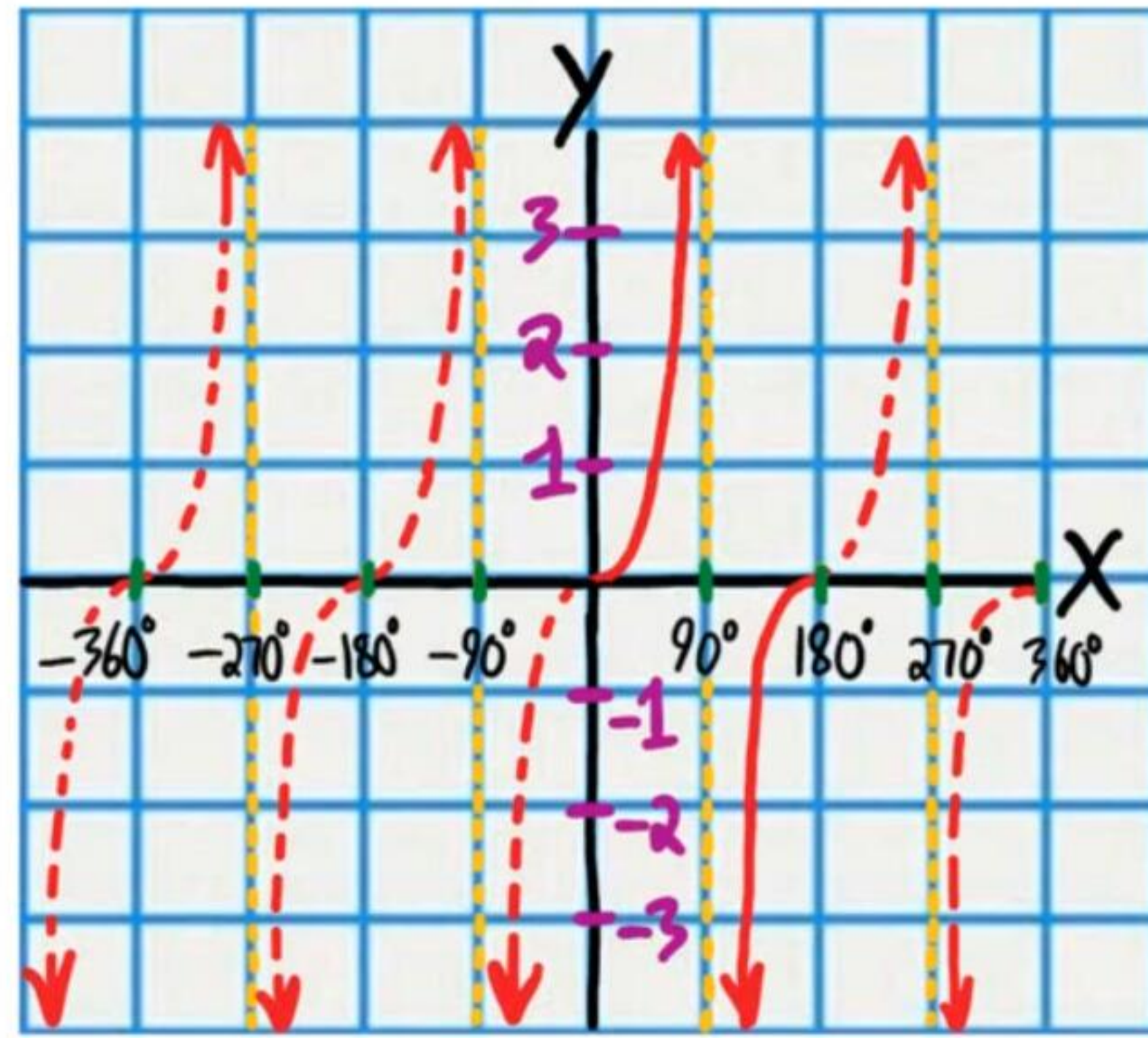

$$y = \sin x$$



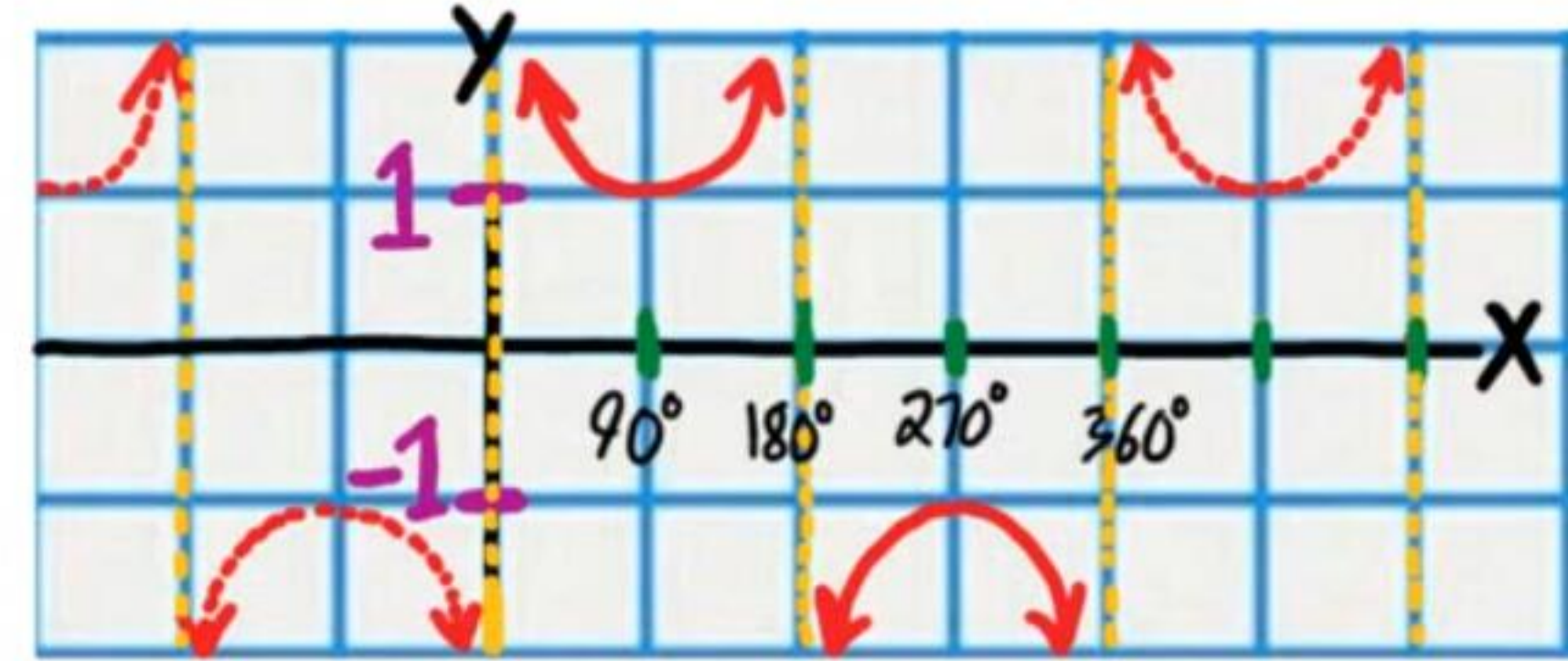
$$y = \cos x$$



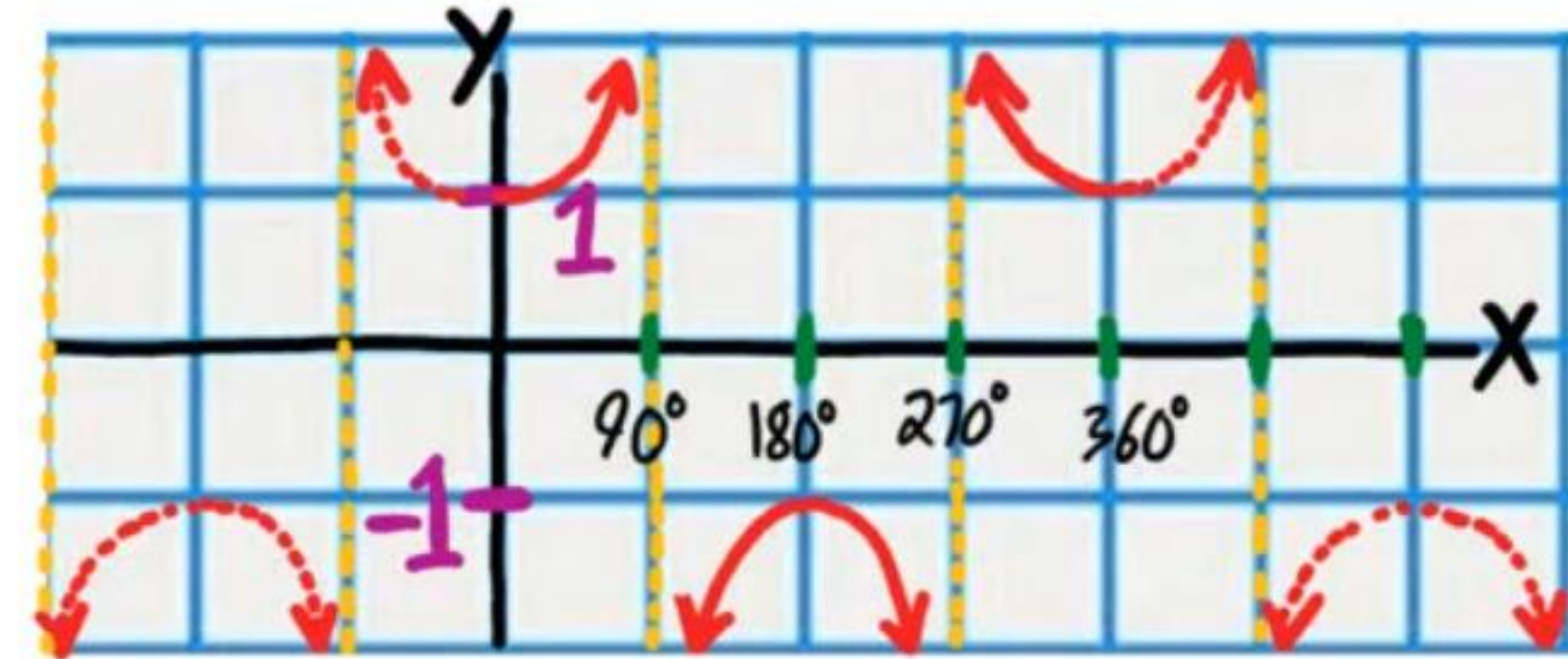
$$y = \tan x$$



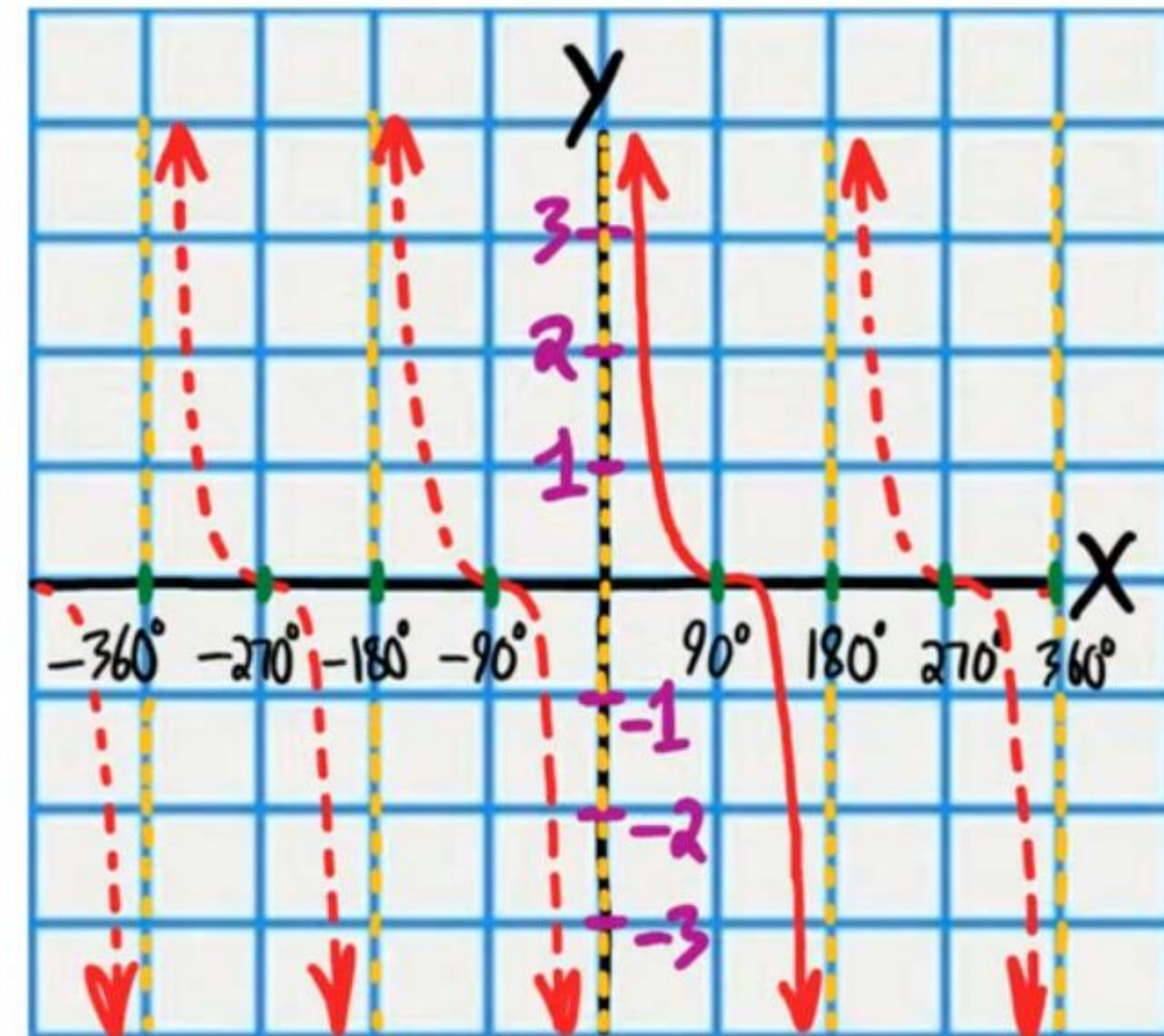
$$y = \csc x$$



$$y = \sec x$$



$$y = \cot x$$



Find the value of the following expression *without a calculator*.

$$\begin{aligned} \textcircled{29} \tan 75^\circ &= \frac{\sin 75^\circ}{\cos 75^\circ} = \frac{\sin(30^\circ + 45^\circ)}{\cos(30^\circ + 45^\circ)} \\ &= \frac{\sin 30^\circ \cos 45^\circ + \sin 45^\circ \cos 30^\circ}{\cos 30^\circ \cos 45^\circ - \sin 30^\circ \sin 45^\circ} \\ &= \frac{\frac{\sqrt{2} + \sqrt{6}}{4} \text{ (4)}}{\frac{\sqrt{6} - \sqrt{2}}{4} \text{ (4)}} \\ &= \frac{\sqrt{2} + \sqrt{6}}{4} \div \frac{\sqrt{6} - \sqrt{2}}{4} \\ &= \frac{\sqrt{2} + \sqrt{6}}{4} \cdot \frac{4}{\sqrt{6} - \sqrt{2}} \\ &= \frac{\sqrt{2} + \sqrt{6}}{\sqrt{6} - \sqrt{2}} \\ &= \frac{(\sqrt{2} + \sqrt{6})(\sqrt{6} + \sqrt{2})}{(\sqrt{6} - \sqrt{2})(\sqrt{6} + \sqrt{2})} \end{aligned}$$

$$= \frac{(\sqrt{2} + \sqrt{6})(\sqrt{6} + \sqrt{2})}{(\sqrt{6} - \sqrt{2})(\sqrt{6} + \sqrt{2})}$$

$$= \frac{\sqrt{12} + \sqrt{4} + \sqrt{36} + \sqrt{12}}{\sqrt{36} + \sqrt{12} - \sqrt{12} - \sqrt{4}}$$

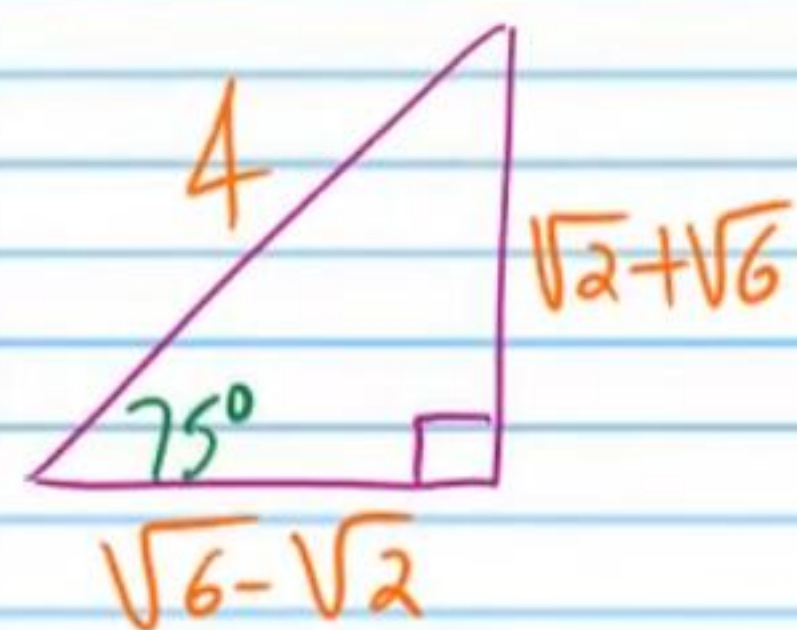
$$= \frac{2\sqrt{3} + 2 + 6 + 2\sqrt{3}}{6 + 0 - 2}$$

$$= \frac{4\sqrt{3} + 8}{4}$$

$$= \frac{4\sqrt{3}}{4} + \frac{8}{4}$$

$$= \boxed{\sqrt{3} + 2}$$

③ Find $\csc 75^\circ$, $\sec 75^\circ$, and $\cot 75^\circ$ without a calculator.



$$\csc 75^\circ = \frac{4}{\sqrt{2} + \sqrt{6}}$$

$$\sec 75^\circ = \frac{4}{\sqrt{6} - \sqrt{2}}$$

$$\cot 75^\circ = \frac{\sqrt{6} - \sqrt{2}}{\sqrt{2} + \sqrt{6}}$$

An Example of Other Identities that Can Be Formed

$$\tan(A+B) = \frac{\sin(A+B)}{\cos(A+B)}$$

$$= \frac{\sin A \cos B + \sin B \cos A}{\cos A \cos B - \sin A \sin B}$$

$$= \frac{\frac{\sin A \cos B + \sin B \cos A}{\cos A \cos B}}{\frac{\cos A \cos B - \sin A \sin B}{\cos A \cos B}}$$

$$= \frac{\frac{\sin A \cos B}{\cos A \cos B} + \frac{\sin B \cos A}{\cos A \cos B}}{\frac{\cos A \cos B - \sin A \sin B}{\cos A \cos B}}$$

$$= \frac{\frac{\sin A \cos B}{\cos A \cos B} + \frac{\sin B \cos A}{\cos A \cos B}}{\frac{\cos A \cos B - \sin A \sin B}{\cos A \cos B}}$$

$$= \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$= \frac{\tan A + \tan B}{1 - \tan A \tan B}$$